

2.5 Precise Definition of a Limit Worksheet

1. Prove that $\lim_{x \rightarrow 3} (4x - 5) = 7$.

Scratch work:

We want to pick a delta such that: “if $0 < |x - 3| < \delta$, then $|4x - 5 - 7| < \varepsilon$ ”. So, we have that $0 < |x - 3| < \delta$. We note that:

$$|4x - 5 - 7| = |4x - 12| = |4(x - 3)| = 4|x - 3| < 4 \cdot \delta$$

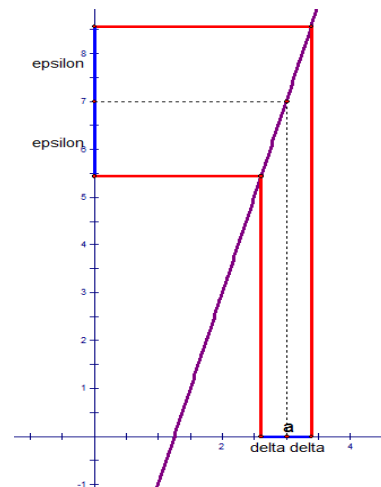
Notice that if we take $\delta = \frac{\varepsilon}{4}$, then $4\delta = \varepsilon$, and we will have that the if-then statement holds.

Solution:

Suppose ε is some positive real number. Let $\delta = \frac{\varepsilon}{4}$, and suppose that $0 < |x - 3| < \delta$.

Then, $|4x - 5 - 7| = |4x - 12| = |4(x - 3)| = 4|x - 3| < 4 \cdot \delta = \varepsilon$

Therefore we can say that $\lim_{x \rightarrow 3} (4x - 5) = 7$.



2. Prove that $\lim_{x \rightarrow \frac{1}{2}} (10x + 6) = 11$.

Scratch work:

We want to pick a delta such that: “if $0 < |x - \frac{1}{2}| < \delta$, then $|10x + 6 - 11| < \varepsilon$ ”. So, we

have that $0 < |x - \frac{1}{2}| < \delta$. We note that:

$$|10x + 6 - 11| = |10x - 5| = \left| 10 \left(x - \frac{1}{2} \right) \right| = 10 \left| x - \frac{1}{2} \right| < 10 \cdot \delta$$

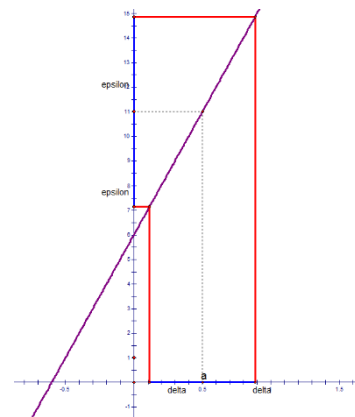
Notice that if we take $\delta = \frac{\varepsilon}{10}$, then $10\delta = \varepsilon$, and we will have that the if-then statement holds.

Solution:

Suppose ε is some positive real number. Let $\delta = \frac{\varepsilon}{10}$, and suppose that $0 < |x - \frac{1}{2}| < \delta$. Then,

$$|10x + 6 - 11| = |10x - 5| = \left| 10 \left(x - \frac{1}{2} \right) \right| = 10 \left| x - \frac{1}{2} \right| < 10 \cdot \delta = \varepsilon$$

Therefore we can say that $\lim_{x \rightarrow \frac{1}{2}} (10x + 6) = 11$.



3. Find $\lim_{x \rightarrow 0} (-2x - 5)$ and prove your answer is correct.

We suspect that $\lim_{x \rightarrow 0} (-2x - 5) = -5$ and now we must prove this is correct.

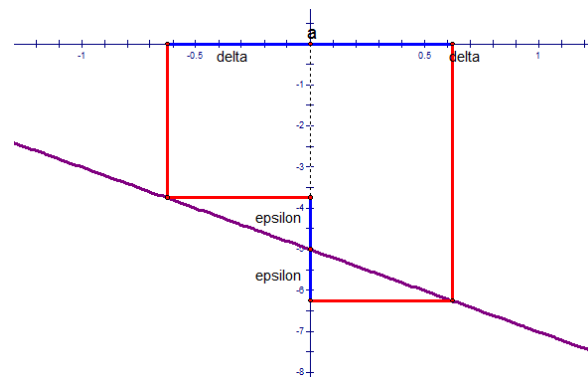
Scratch work:

We want to pick a delta such that: “if $0 < |x - 0| < \delta$, then

$|-2x - 5 - (-5)| < \varepsilon$ ”. So, we have that $0 < |x| < \delta$. We note that:

$$|-2x - 5 - (-5)| = |-2x| = |2x| = 2|x| < 2 \cdot \delta$$

Notice that if we take $\delta = \frac{\varepsilon}{2}$, then $2\delta = \varepsilon$, and we will have that the if-then statement holds.



Solution:

Suppose ε is some positive real number. Let $\delta = \frac{\varepsilon}{2}$, and suppose that $0 < |x| < \delta$. Then,

$$|-2x - 5 - (-5)| = |-2x| = |2x| = 2|x| < 2 \cdot \delta = \varepsilon$$

Therefore we can say that $\lim_{x \rightarrow 0} (-2x - 5) = -5$.

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4. Find $\lim_{x \rightarrow \pi} \left(\frac{2}{\pi} x - e \right)$ and prove your answer is correct.

We suspect that $\lim_{x \rightarrow \pi} \left(\frac{2}{\pi} x - e \right) = 2 - e$ and now we must prove this is correct.

Scratch work:

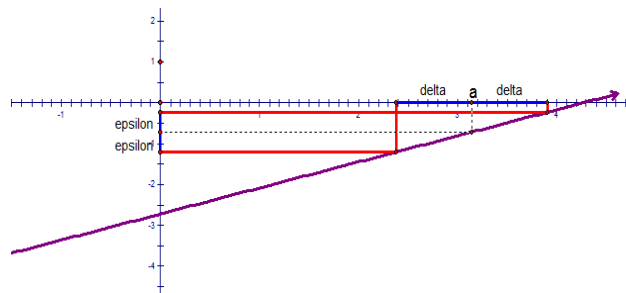
We want to pick a delta such that: “if $0 < |x - \pi| < \delta$, then

$\left| \frac{2}{\pi} x - e - (2 - e) \right| < \varepsilon$ ”. So, we have that $0 < |x - \pi| < \delta$. We note

that:

$$\left| \frac{2}{\pi} x - e - (2 - e) \right| = \left| \frac{2}{\pi} x - 2 \right| = \frac{2}{\pi} |x - \pi| < \frac{2}{\pi} \cdot \delta$$

Notice that if we take $\delta = \frac{\pi}{2} \varepsilon$, then $\frac{2}{\pi} \delta = \varepsilon$, and we will have that the if-then statement holds.



Solution:

Suppose ε is some positive real number. Let $\delta = \frac{\pi}{2} \varepsilon$, and suppose that $0 < |x - \pi| < \delta$. Then,

$$\left| \frac{2}{\pi} x - e - (2 - e) \right| = \left| \frac{2}{\pi} x - 2 \right| = \frac{2}{\pi} |x - \pi| < \frac{2}{\pi} \cdot \delta = \varepsilon$$

Therefore we can say that $\lim_{x \rightarrow \pi} \left(\frac{2}{\pi} x - e \right) = 2 - e$.

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